Boyuan Yu

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Education

PhD degree in Civil Engineering — *McGill University*
PhD degree program — *September 2020 - October 2024*
CGPA: 4.00/4.00.
Thesis title: Modelling and Simulation of Extreme Waves in Floods and Mudflows.

Master's degree in Civil Engineering — *McGill University*
Master's degree program — *September 2018 - May 2020*
CGPA: 3.96/4.00.
Thesis title: Transverse shear instability in steep open-channel flow, Link.

Bachelor's degree in Water Conservancy and Hydropower Engineering — *Hohai University*
Bachelor's degree program — *September 2014 - May 2018*
CGPA: 4.80/5.00.
Ranking: 1/155.

Working Experience

Department of Civil Engineering — *McGill University*
Research assistant — *October 2024 - December 2024*

Department of Civil Engineering — *McGill University*
Postdoc researcher — *January 2025 - Now*

Citizenship Status

People's Republic of China

Peer-Reviewed Publications and Research Projects

A. Journal papers

AC: corresponding author.

1. **Boyuan Yu**, and Vincent H. Chu,
   The front runner in roll waves produced by local disturbances,
   Journal of Fluid Mechanics, 932, A42 (2022) [18 pages];
   DOI: 10.1017/jfm.2021.1011.
2. **Boyuan Yu**, and Vincent H. Chu,
   Impact force of roll waves against obstacles,
   Journal of Fluid Mechanics, 969, A31 (2023) [25 pages];
   DOI: 10.1017/jfm.2023.580.
3. **Boyuan Yu (AC)**, and Vincent H. Chu,
   Roll Waves in Mudflow Modelled as Herschel-Bulkley Fluids,
   Journal of Engineering Mechanics, (2024) [17 pages];
   DOI: 10.1061/JENMDT/EMENG-7931.

4. **Boyuan Yu**, and Vincent H. Chu,
   Roll Waves on a Laminar Sheet Flow of Newtonian Fluid with Negligible Surface Tension,
   Journal of Fluid Mechanics, 999, A49 , (2024) [31 pages];
   DOI: 10.1017/jfm.2024.887.

5. **Boyuan Yu**, and Vincent H. Chu,
   Direct Numerical Simulation of Roll Waves on Landslide Mudflow,
   Physics of Fluids, 36 (12) (2024) [24 two-column manuscript pages];
   DOI: 10.1063/5.0239285.

6. **Boyuan Yu (AC)**
   Improved prediction of the front runner in roll waves produced by localized disturbance,
   Wave Motion, 134(3):103484 (2025) [17 pages];
   DOI: 10.1016/j.wavemoti.2024.103484.

7. **Boyuan Yu (AC)**,
   Two-layer Roll Waves in Rapid Gravity Currents on Mild Slopes,
   Wave Motion (2025) [25 pages];
   DOI: 10.1016/j.wavemoti.2025.103634.

8. **Boyuan Yu (AC)**, Shinto Roose, Laxmi Sushama, Teufel Bernardo, and Ram Yerubandi,
   High water risks for the Great Slave Lake Shoreline,
   Accepted by Results in Engineering.

B. Conference papers

1. **Boyuan Yu**, and Vincent H. Chu,
   Wave and bed-friction effect on instability of shear flow in shallow waters,
   the 10th Conference on Fluvial Hydraulics - River Flow 2020: Delft, Netherlands [8 pages];
   DOI: 10.1201/b22619-12.

2. **Boyuan Yu**, and Vincent H. Chu,
   Impact force of the roll waves produced by local disturbances,
   the 39th IAHR World Congress (2022): Granada, Spain [10 pages];
   DOI: 10.3850/IAHR-39WC2521711920221273.

3. **Boyuan Yu**, and Vincent H. Chu,
   Roll Waves on a Laminar Sheet Flow produced by Local Disturbance,
   the 11th International Conference on Fluvial Hydraulics - River Flow 2022: Kingston and Ottawa, Canada [8 pages];
   DOI: 10.1201/9781003323037-79.

4. **Boyuan Yu**, and Vincent H. Chu,
   Impact of Mud Flow Instabilities on Hydraulic Structures,
   the 11th International Conference on Fluvial Hydraulics - River Flow 2022: Kingston and Ottawa, Canada [9 pages];
   DOI: 10.1201/9781003323037-80.

5. **Boyuan Yu**, and Vincent H. Chu,
   The Impact of Flood Waves on Hydraulic Structures,
   the 11th International Conference on Fluvial Hydraulics - River Flow 2022: Kingston and Ottawa, Canada [8 pages];
   DOI: 10.1201/9781003323037-140.

6. **Boyuan Yu**, and Vincent H. Chu,
   The Front Runner of Roll Wave in Mudflow,
   Accpted by Proceedings of the ASME 2024 43nd International Conference on Ocean, Offshore and

Arctic Engineering OMAE2024: Singapore [10 pages];
DOI: 10.1115/OMAE2024-121514.

7. **Boyuan Yu**, and Vincent H. Chu,
   Impact of Roll Waves in Mudflow on Hydraulic Structures,
   Accpted by Proceedings of the ASME 2024 43nd International Conference on Ocean, Offshore and Arctic Engineering OMAE2024: Singapore [10 pages];
   DOI: 10.1115/OMAE2024-121513.

8. **Boyuan Yu**, and Vincent H. Chu,
   The Front Runner of Roll Waves in Jiang-Jia Ravine,
   Accpted by the 10th International Symposium on Hydraulic Structures 2024: Zurich, Switzerland [9 pages, withdrew due to visa reasons].

9. **Boyuan Yu**, and Vincent H. Chu,
   Roll Waves on Landslide Mudflow against Structures of Various Shapes and Orientations,
   Accpted by the 10th International Symposium on Hydraulic Structures 2024: Zurich, Switzerland [10 pages, withdrew due to visa reasons].

C. Other Finished and Ongoing Research

1. **Boyuan Yu**, Ge Gao
   Impact of Free-surface Viscoelastic Fluid against Obstacles on Mildly Inclined Planes,
   In submission to Journal of Non-Newtonian Fluid Mechanics.

2. **Boyuan Yu**, Oveys Ziya, and Laxmi Sushama,
   Human Instability Assessment under Critical Extreme Urban Flooding Considering the Topography Effect,
   In preparation (ready for submission).

3. **Boyuan Yu**, and Laxmi Sushama,
   River ice process and clustering in Mackenzie Basin,
   In preparation.

4. **Boyuan Yu**,
   Roll wave development in collapsible tubes modelled using 1D and 3D models,
   In preparation.

5. **Boyuan Yu**, and Vincent H. Chu,
   Impact Force of Roll Waves on Mudflow Modelled as Power-law Fluids,
   In preparation.

Talks and Presentations

1. **Boyuan Yu**, and Vincent H. Chu,
   The video animation related to the conference paper *Impact of Mud Flow Instabilities on Hydraulic Structures*,
   River Flow 2022 Conference Best Video Contest, 2022.

2. **Boyuan Yu**,
   Modelling and Simulation of Roll Waves Using Basilisk and Gerris, Link,
   Basilisk Monthly Meeting, October 29th, 2024.

Awards

1. **Boyuan Yu**,
   National Scholarship by the Ministry of Education of the PRC (top 1% of the undergraduate students every year), 2015.

2. **Boyuan Yu**,
   Excellent Graduate Student Award by Hohai University (top 5% of the graduate student), 2018.
3. **Boyuan Yu**,
   Grad Excellence Award in Civil Engineering & Applied Mechanics by McGill University, 2019.
4. **Boyuan Yu**,
   MEDA (McGill Engineering Doctoral Award) scholarship by McGill University, 2020.
5. **Boyuan Yu**, and Vincent H. Chu,
   OMAE2024 Best Paper Award for conference paper OMAE2024-121514, 2024.

Research Projects

1. Grant number: RGPIN-2019-05776. Role: Researcher. Funding Body: The Natural Sciences and Engineering Research Council of Canada. Duration: 01/09/2020–04/10/2024. Funding Received: CAD $55,500.

Teaching and Mentoring

Teaching Assistant — *September 2019 - May 2024*
McGill University — *Montreal, Canada*

- CIVE 281: Analytical Mechanics.
- CIVE 327: Fluid Mechanics and Hydraulics.
- CIVE 572: Computational Hydraulics.

Research Interests

- Hydrodynamic instabilities.
- Shallow water equations.
- Hyperbolic conservation laws.
- Finite volume method.
- Riemann solvers.
- Roll waves and shear instabilities.
- Multiphase flow.
- Non-Newtonian fluids.
  - Visco-plastic fluids.
  - Visco-elastic fluids.
  - Granular flow.
- Wave impact forces on structures.
- Open-source CFD.
- Multilayer model.
- Spectral method.
- Stratified flow, internal waves.
- Flood and storm surge.

- River ice, lake ice and ocean ice.

Technical Skills

Programming Languages/Tools	Matlab, Mathematica, Python, C, Fortran, Julia
Numerical models for CFD	Basilisk, Gerris, OpenFOAM, Clawpack, PyPDE, Centpy, Wavemaker, Mike Zero, ADCIRC, Schism, River1D, Ansys, DualSPHysics, Open Telemac, Star-CCM+, Flow-3D
Postprocessing tools for CFD	Tecplot, Paraview, OriginLab, Microsoft Visio, Gnuplot, Qgis, SMS, Blender
Text processing	LaTeX, Microsoft Word, Markdown & Obsidian
Operating system	Windows, Linux
Video editing	Kdenlive
Languages	Mandarin (mother tongue), English (proficient), Italian (elementary), Cantonese (beginner), French (beginner), Spanish (beginner)